PLACIFY

AI-Augmented Placement Preparation & Career Networking Platform

1. Abstract

Placify is an AI-augmented placement preparation and career networking platform designed for students and working professionals. It centralises the entire placement journey from community-driven knowledge sharing and peer networking to intelligent resume analysis, into a single cohesive web application.

By integrating a Large Language Model (LLM) powered ATS (Applicant Tracking System) analyser with real-time link verification, Placify bridges the gap between raw placement guidance and actionable, personalised career insights.

2. Platform Description

Placify is a role-aware, full-stack web application that serves three distinct user personas:

- Students (Unplaced) — Seeking guidance, resume help, job opportunities, and community support.
- Students (Placed) — Sharing experiences, strategies, and sample resumes with peers.
- Working Professionals (HR / Employee) — Posting opportunities, engaging with the community, and offering mentorship.

After registration, users complete a personalised onboarding flow that captures their profile type and role, which governs the features and content they see. The platform combines a real-time community forum, an AI-powered ATS resume scanner, and a resume builder module all behind a secure, JWT-authenticated backend.

3. Problem Statement

The campus placement process is fragmented and opaque for most students. The table below summarises the core problems Placify addresses:

Problem	Impact
Lack of centralised placement insights	Students rely on scattered WhatsApp groups and word-of-mouth
No feedback on resume quality before applying	Resumes get silently rejected by ATS systems

Problem	Impact
Broken / fabricated links on resumes go undetected	Recruiters distrust candidate profiles
Placed students have no structured channel to give back	Knowledge transfer is lost each batch

Placify solves these challenges by providing:

- A structured community forum categorised by placement topics.
- An AI-powered ATS analyser that scores resumes, flags formatting issues, and suggests improvements.
- A link verification engine that independently validates every URL in a resume before the AI evaluates it.
- A placed-students repository where verified past candidates can share winning resumes and strategies.

4. System Architecture

Placify follows a three-tier architecture: a React single-page application (SPA) on the frontend, an Express.js REST API on the backend following the MVC pattern, and a MongoDB database managed via Mongoose ODM.

4.1 Client Layer (React.js)

The client is a Create React App project styled with Tailwind CSS. Routing is handled by React Router DOM v6. The application flow is:

- Landing Page → Register / Login → Onboarding → Dashboard

The Dashboard exposes four main sections based on user role:

Section	Status	Description
Community (Forum)	Live	Real-time post feed with categories, likes, and comments
ATS Analyser	Live	AI + link-verifier powered resume scoring
Resume Builder	Pending	Interactive builder with ATS-safe templates
Chat	Pending	One-on-one real-time messaging via WebSocket

4.2 Backend Layer (Express.js — MVC Architecture)

The backend is an Express.js 5 application structured using the Model-View-Controller pattern. Key modules include:

- Auth Layer — Register, Login, Onboarding, JWT issuance, Profile management.

- Posts / Forum — CRUD posts, Like / Unlike, nested comments, pagination, category filtering.
- ATS Module — File upload, text extraction, link verification, LLM call, structured JSON response.
- JWT Middleware — Auth guard protecting all private routes.
- Global Error Handler — notFound and errorHandler middleware.
- Health Check — GET /api/health endpoint for uptime monitoring.

4.3 Database Layer (MongoDB)

MongoDB serves as the primary NoSQL document store, accessed via the Mongoose ODM. The following collections are defined:

- Users — account credentials, profile type, role, LinkedIn URL.
- Posts — forum posts with category, author ref, likes array, and comment count.
- Comments — nested comments linked to posts and authors.

4.4 External Integrations

Service	Purpose
Groq Cloud (LLM Inference)	Hosts the llama-3.3-70b-versatile model for ATS analysis
GitHub REST API v3	Fetches live repository metadata (stars, language, last push, visibility)
HTTP HEAD Checks	Validates all non-GitHub URLs (LinkedIn, certifications, portfolio links)

4.5 Request Lifecycle

A complete end-to-end request through the ATS Analyser follows these steps:

- User uploads resume (PDF or DOCX) via multipart/form-data POST to /api/ats/analyze.
- Backend extracts raw text using pdf-parse or Mammoth.
- Link Verifier extracts all URLs (plain-text regex + PDF annotation scan + DOCX .rels XML).
- Each URL is verified — GitHub repos via the GitHub API, all others via HTTP HEAD.
- Resume text, optional job description, and link verification summary are sent to Groq LLM.
- Structured JSON is returned and rendered in the Dashboard ATS panel.

5. Technology Stack & Dependencies

5.1 Frontend

Category	Technology	Version
UI Framework	React	18.2.0
Routing	React Router DOM	6.30.3
Styling	Tailwind CSS	3.4.19
Icons	Lucide React	0.575.0
Build Tool	Create React App (react-scripts)	5.0.1
PostCSS	Autoprefixer	10.4.27

5.2 Backend

Category	Technology	Version
Runtime	Node.js	LTS
Web Framework	Express	5.2.1
Database ODM	Mongoose	8.18.0
Authentication	JSON Web Token (jsonwebtoken)	9.0.2
Password Hashing	bcryptjs	2.4.3
File Uploads	Multer	2.1.1
PDF Parsing	pdf-parse	1.1.1
DOCX Parsing	Mammoth	1.11.0
ZIP Handling (DOCX)	JSZip	3.10.1
Cross-Origin	cors	2.8.6
Environment Config	dotenv	16.4.5
AI SDK	groq-sdk	0.37.0
Dev Server	nodemon	3.1.7

5.3 Database

Technology	Role
MongoDB	Primary NoSQL document database
Mongoose	Schema definition, validation, and ODM layer

6. AI Stack

6.1 Model Configuration

Component	Details
Provider	Groq Cloud — ultra-fast LLM inference
Model	llama-3.3-70b-versatile (70B parameter Meta LLaMA 3.3)
Temperature	0.2 (deterministic, low creativity for structured output)
Max Tokens	3,000 per analysis response

6.2 ATS System Prompt Design

The AI is given a carefully engineered system prompt that instructs it to:

- Evaluate the resume against ATS criteria using a 7-section rubric: header, summary, experience, education, skills, certifications, and projects.
- Perform keyword matching against the provided job description, or infer the role if none is provided.
- Detect formatting issues (tables, graphics, columns, non-standard fonts) that break ATS parsers.
- Return a strict JSON-only output (no markdown preamble) with fields: score, scoreRationale, detectedRole, sectionNotes, keywordMatch, formattingIssues, improvementChecklist, and nextSteps.

6.3 Link Verification Agent

A custom rule-based verification agent (linkVerifier.js) runs before every LLM call, ensuring the AI analyses only verified, real-world link data:

Method	Description
Plain-text regex	Extracts all http(s):// URLs from the raw resume text
PDF binary scan	Parses /URI annotation entries from the PDF byte stream (3 patterns: parenthesis, hex-encoded, /A << style)
DOCX relationship XML	Unzips the DOCX archive and parses word/**/*.rels XML files for hyperlink entries
GitHub API check	Calls api.github.com/repos/{owner}/{repo} to get live metadata (stars, language, last push, visibility)
HTTP HEAD check	Fires HTTP HEAD for all other URLs; HTTP 2xx–3xx and 403/405 (bot-blocked live servers) are treated as live

7. Features Implemented

7.1 Authentication & User Management

- ✓ User registration with email + password (bcrypt hashed)
- ✓ JWT-based login with 7-day token expiry
- ✓ Protected routes via authenticate middleware
- ✓ Role-aware onboarding (Student: Placed / Unplaced, Working Professional: HR / Employee)
- ✓ LinkedIn profile URL stored on user profile
- ✓ User search / directory (GET /api/auth/users?search=)

7.2 Landing Page & UI

- ✓ Full marketing landing page (Hero, Problem Section, How It Works, For Senior Section, Social Proof, CTA, Footer)
- ✓ Responsive Navbar with authentication state awareness
- ✓ Smooth scroll and animated components throughout

7.3 Dashboard

- ✓ Role-aware sidebar navigation
- ✓ Dashboard home with personalised stats and quick-access cards
- ✓ User profile section (view and update LinkedIn URL)

7.4 Community Forum

- ✓ Create, read, and delete posts
- ✓ Category filtering: All Posts, Internships, Placements, DSA, Aptitude, Interview Experience, and more
- ✓ Paginated post feed (20 posts per page)
- ✓ Like / Unlike posts (toggle)
- ✓ Nested comments (add and delete own comments)
- ✓ Comment count displayed on post cards

7.5 ATS Resume Analyser

- ✓ Resume upload (PDF and DOCX supported)
- ✓ Optional job description input for targeted analysis
- ✓ AI-powered ATS scoring (0–100) with detailed rationale

- ✓ Section-by-section analysis: header, summary, experience, education, skills, certifications, projects
- ✓ Keyword match report (matched, partial, missing, density)
- ✓ Formatting issue detection with severity levels: high / medium / low
- ✓ Prioritised improvement checklist with before / after examples
- ✓ Next steps summary panel
- ✓ Link verification panel showing live / dead status for GitHub repos, LinkedIn, and certifications
- ✓ GitHub repository metadata display: stars, language, last push date, visibility

7.6 Backend Infrastructure

- ✓ MVC-structured Express backend
- ✓ Global error handler middleware (notFound and errorHandler)
- ✓ CORS policy with configurable CLIENT_ORIGIN environment variable
- ✓ Health check endpoint: GET /api/health
- ✓ Environment variable management with dotenv

8. Features Roadmap (To Be Implemented)

8.1 Resume Builder

- ◻ Interactive resume builder with section-by-section editor
- ◻ Multiple resume templates with ATS-safe designs
- ◻ PDF export of the generated resume
- ◻ Auto spell-check and grammar correction

8.2 Chat / Messaging

- ◻ Real-time one-on-one chat between users via WebSocket / Socket.IO
- ◻ Message history persistence in MongoDB
- ◻ Online presence and typing indicators

8.3 Placed Resumes Showcase

- ◻ Placed students can upload anonymised winning resumes
- ◻ Category-tagged by company, role, year, and package range
- ◻ Search and filter placed resume repository
- ◻ Download with access control (login required)

8.4 AI Enhancements

- ◻ AI-powered cover letter generator based on resume and job description
- ◻ Streaming ATS analysis output (real-time token-by-token response)
- ◻ Mock interview Q&A generator with role-specific questions from job description
- ◻ Smart job-description matcher: upload multiple JDs and rank by resume fit

8.5 Community & Social

- ◻ Post bookmarks / save for later
- ◻ Rich text editor for posts (Markdown / WYSIWYG)
- ◻ Upvote / downvote system for comments
- ◻ User follow system and personalised feed
- ◻ Notification system for new comments, likes, and mentions

8.6 Platform & DevOps

- ◻ Admin dashboard for user management and post moderation
- ◻ Email verification on registration
- ◻ Password reset via email (OTP / magic link)
- ◻ Rate limiting and request throttling
- ◻ Deployment pipeline (CI/CD) to Render + Vercel / MongoDB Atlas
- ◻ Unit and integration tests using Jest and Supertest

9. Conclusion

Placify addresses a real and pressing gap in the campus placement ecosystem by building a unified, intelligent platform that serves students and professionals alike. The current implementation establishes a robust full-stack foundation: a secure JWT-authenticated Express backend following the MVC architectural pattern, a MongoDB data layer with well-structured Mongoose schemas, and a React frontend with Tailwind CSS styling.

The standout feature at this stage is the AI-powered ATS Resume Analyser, which goes beyond a simple LLM call by incorporating a multi-source link verification agent that independently validates every URL embedded in the resume from GitHub repositories (fetching live metadata via the GitHub API) to LinkedIn profiles and certification badges and feeds that verified context into the Groq llama-3.3-70b-versatile model for a more accurate, evidence-grounded analysis.

With a clear roadmap of features to implement real-time chat, resume building, AI cover letter generation, and a placed-students repository Placify is engineered to grow into a comprehensive career platform that empowers students to enter the workforce confidently and well-prepared.